Year 12 Maths - Applications of differentiation

Similar questions - Stationary points and inflection

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A curve has equation $y = 2x^3 - 9x^2 + 12x + k$, where k is a constant.

(a) Find the coordinates, in terms of k, of the two stationary points and determine their nature. **[6]**

(b) Hence find the range of values of k for which the equation $2x^3 - 9x^2 + 12x + k = 0$ has exactly three distinct real roots. **[3]**

(c) Show that the equation has exactly one real root when $k = -3$. **[2]**

2 A curve has equation $y = x^3 + 3x^2 + 4x$.

(a) Show that the curve has no stationary points. **[3]**

(b) Show that the curve has exactly one point of inflection, and find its coordinates. **[3]**

(c) Find the least gradient of the curve, and the point at which it occurs. **[2]**

Question 1

(a) $\dfrac{dy}{dx} = 6x^2 - 18x + 12 = 6(x-1)(x-2)$. Stationary points at $(1, 5+k)$ and $(2, 4+k)$. $\dfrac{d^2y}{dx^2} = 12x - 18$: at $x = 1$ it is $-6 < 0$ (maximum); at $x = 2$ it is $6 > 0$ (minimum)

(b) Three distinct roots need the maximum above and the minimum below the x-axis: $5 + k > 0$ and $4 + k < 0$, so $-5 < k < -4$

(c) With $k = -3$ the minimum is at $y = 1 > 0$ and the maximum at $y = 2 > 0$, so the curve crosses the x-axis only once (to the left of the maximum)

Question 2

(a) $\dfrac{dy}{dx} = 3x^2 + 6x + 4$ has discriminant $36 - 48 = -12 < 0$, so it is never zero

(b) $\dfrac{d^2y}{dx^2} = 6x + 6$, zero only at $x = -1$ and changes sign there. $y = -1 + 3 - 4 = -2$: $(-1, -2)$

(c) $\dfrac{dy}{dx} = 3(x+1)^2 + 1 \geqslant 1$, with least value 1 at $x = -1$, the point $(-1, -2)$